

Harsh Mishra

B.Tech - Information Technology
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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech.	Dr. A.P.J. Abdul Kalam Technical University,Lucknow	7.48	2020-2024

WORK EXPERIENCE

Probe42

Software Engineer Intern

Jul 2024 - present

Bengaluru, India

- Image Processing Solution: Built using **OpenCV** and Tesseract for **OCR**, enhancing document readiness.
- Implemented containerized OCR pipelines using **Docker** and **Kubernetes**, improving scalability and processing efficiency.
- Increased OCR Efficiency: Improved OCR efficiency from 89.4 percent to 90.7 percent, optimizing the overall accuracy of data extraction.
- Built a dataset and fine-tuned **LayoutLMv3** model, utilizing **PyTorch** for training ,extraction and labeling.
- Enhanced monitoring and logging with AWS CloudWatch, Prometheus and Graphana for real-time performance tracking.
- MIS Captcha Monitoring System**: Developed a Management Information System (MIS) application for CAPTCHA visualization and performance monitoring of automated solving systems.
- Designed and implemented full-stack application using **Angular** for Frontend visualization and **Flask** for Backend API development
- Developed **RESTful APIs** for data ingestion from distributed CAPTCHA solving nodes and real-time dashboard updates
- Implemented data aggregation pipelines for calculating solving accuracy (98.2%), average response time (1.4s), and system uptime (99.95%)
- Integrated **RabbitMQ** as a message broker to handle asynchronous task processing between microservices, reducing response latency and improving system scalability.

PROJECTS

Cycle Cruise

Personal Project

Apr 2023 - Jun 2023

Github

- Engineered a responsive cycle rental web application, carefully designing a user-friendly platform for effortless cycle booking on a daily basis.
- Utilized the MERN stack (MongoDB, Express.js, ReactJS, Node.js) and Redux for state management to create a seamless user experience.
- Deployed application using Docker and Kubernetes for scalability, utilizing AWS EC2 and S3 for hosting and storage.
- Implemented CI/CD pipelines using GitHub Actions to automate deployment and testing.

Text Summarizer

Personal Project

Oct 2022 - Nov 2022

Github

- Developed a machine learning model that takes a blog as input and generates a coherent summary as output.
- Containerized the application using Docker for easy deployment and scalability.
- Aimed to produce a fluent summary highlighting the main points of the document, achieving an accuracy of 93 percent.
- Managed infrastructure using Terraform and automated deployment using Ansible.
- Deployed on **AWS Lambda** for serverless execution.
- Utilized libraries including **NumPy**, **Pandas**, NLTK and Scikit-learn for implementation.

TECHNICAL SKILLS

- Languages**: C, C++, Python, TypeScript, Java, HTML, Bash, CSS
- Frameworks**: Angular, NodeJs, ReactJs, ExpressJs, Pytorch, Flask
- Databases**: MySQL, Mongoddb, AWS S3
- Miscellaneous**: AWS Lambda, AWS S3, EC2, CloudWatch, Graphana, Prometheus, Docker, Kubernetes, BitBucket.

KEY COURSES TAKEN

- Data Structures and Algorithms
- Operating Systems
- Database Management System
- Data Analytics and Machine Learning
- Image Processing
- Object-Oriented Programming
- Computer Networks
- Cloud Computing

ACHIEVEMENTS

- Top 100**, in AICTE presents Innovate India Coding Championship (IICC) in association with Coding Ninjas and Chandigarh University among 1 lakh+ students.
- SIH finalist**, Smart India Hackathon 2022 finalist
- 500+ Questions**, Solved over 400 questions on different platforms like Leetcode, CodeStudio, GeeksForGeeks, etc.